A Comparative Evaluation of Elo Ratings- and Machine Learning-based Methods for Tennis Match Result Prediction

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Introduction

Tennis is among the most popular sports in the world, as demonstrated by its levels of participation and spectatorship.[1] According to the International Tennis Federation's 2019 Global Tennis Report,[2] 87 million people (1.17% of the world's population) play tennis worldwide and, although figures are difficult to find in academic literature, tennis consistently appears in the top ten most popular global sports in online lists in terms of both participation and spectatorship. Due to the sport's popularity, there is great interest in forecasting the results of matches, for example, from fans and bookmaking companies. Bettors are interested in match result prediction to decide whether to place bets on matches. Bookmakers use predictive models as well as betting volumes to ensure their match betting odds are appropriate. Coaches and sport performance analysts are interested in interpretable models that allow them to identify the most important features that have contributed to wins in the past, or that players can alter in the future to improve their chances of winning.

Approaches to forecast match results in tennis include ratings-, regression-, points-, and paired comparison-based methods,[3] as well as machine learning models. Ratings-based methods use the difference in player ratings to identify the probable winner of a match, with ratings updated over time based on actual match results. Machine learning (ML) models are trained on a specific number of historical matches and are then tested on an unseen test set to confirm their generalizability. ML models belong to two broad groups: classification models and regression models, which aim to predict discrete and numeric target variables, respectively. In classification, which is considered in the current study, a set of input features is used to predict a labelled binary outcome variable (for example, win or lose). In regression, the target variable might be the margin-of-victory in terms of sets won, that is, the number of sets won by one player minus the number of sets won by the opposition player in a particular match[4] (Games won can also be used as the margin-of-victory; however, a player can win a match but have won fewer games than the opposition).

Elo ratings, which were originally used for the analysis of chess players,[5] have been commonly used to estimate team ratings and predict outcomes in various sports including Football, Australian Rules Football, and Gaelic Football.[6–12] Elo has been extended, for example, to incorporate home advantage,[7] margin-of-victory,[4,6] form/momentum,[13] and different playing surfaces, for example, grass, hard and clay courts.[14] Kovalchik[3] compared the performance of several models in predicting 2,395 Association of Tennis Professionals (ATP) tennis matches in 2014 and found that Elo performed better than regression-, points- and paired comparison-based models. Weighted Elo (WElo), recently proposed by Angelini et al.[13], extends standard Elo to consider form by incorporating a player's most recent match result. WElo was applied to over 60,000 ATP and Women's Tennis Association (WTA) matches, and it was found to outperform paired comparisons, logistic and probit regression, and standard Elo.[13] However, whether Elo and/or WElo can outperform common ML models, for example, those used in a recent study by Wilkens:[15] boosted trees, random forests (RFs), support vector machines (SVMs), and artificial neural networks (ANNs), is an open question. In this study, a comparative evaluation of Elo/WElo against ML models is conducted using the same ATP match data as Angelini et al.[13]

Alternating decision trees (ADTrees/ADTs)[16,17] have been applied in various domains such as natural language processing,[18] medicine,[19,20] and flooding/land-slide susceptibility modeling and assessment,[21–23] but have not yet been applied in the context of sports. ADTrees are used as the boosting method in this study.

This study has three main contributions. First, a comparative evaluation of Elo/WElo with ML models is conducted, which is useful for researchers and practitioners in sports match result forecasting. Second, ADTrees are introduced in sports match result forecasting, which, to our knowledge, have not previously been applied for this purpose. Third, the previously proposed SRP-CRISP-DM[24] framework is demonstrated in practice.

The remainder of this paper is organized as follows. In the next section, background is provided on studies that have used ML or ratings to predict tennis match results. Then, Elo and WElo are described, as is how the SRP-CRISP-DM framework will be applied. The results are then presented and discussed before concluding the paper.

Background

In this section, first, studies that have used ML models for tennis match result prediction are reviewed, followed by studies that have used paired comparison/rating methods.

Machine Learning for Tennis Match Result Prediction

Logistic regression was commonly applied in early studies. Clarke and Dyte[25] used data from the Men's 1998 Wimbledon, 1998 US Open, and 1999 Australian Open Grand Slam tournaments, and fitted a logistic regression model to official ATP rankings to estimate a player's chance of winning as a function of the rating points difference. Klaassen and Magnus[26] applied logistic regression at both match- and point-levels to forecast the match winner at the beginning of, and during, Wimbledon grand slam matches. The estimated win probabilities were functions of the player ranking difference. Lisi[27] used logistic regression to predict 501 tennis matches using ATP points/rankings, the players' ages, home advantage, and information derived from bookmaker odds. The model was estimated using tournament data from 2012 and was applied to Grand Slam tournament matches from 2013, and the model-implied odds were used to place out-of-sample bets. Makino, Odaka and Kuroiwa[28] calculated the frequency of single shots, two-shot patterns, and effective shot patterns from individual points in ATP matches. An L1-regularized logistic regression was then used to predict the winners of points and to obtain useful tactical features. The highest accuracy achieved was 66.5%, and the most relevant features were the respective ratios of shots played by each player.

Recently, ML models other than logistic regression have been applied for tennis match result prediction. Ghosh, Sahu and Biswas[29] compared the performance of a decision tree, learning vector quantization, and an SVM to predict tennis match results, using eight UCI databases of Grand Slam tournament results. The decision tree was found to perform best, achieving 99.14% and 98.45% accuracy for a 70%/30% training/test split and 10-fold cross-validation, respectively. The very high accuracy reported in this study relative to others may indicate that future matches were incorrectly used to predict past matches. Gu and Saaty[30] predicted tennis match results using data and judgments. An analytic network process[31] that incorporated factor analysis and clustering was applied to 63 men's and 31 women's 2015 US Open matches, and achieved 85.1% accuracy. Candila and Palazzo[32] aimed for accurate predictions and profitable bets on men's tennis matches, applying an ANN to a multitude of features. When predicting out-of-sample, the ANN outperformed four of the five other models, regardless of the time period considered. Gao and Kowalczyk[33] used a random forest model to predict tennis match results based on player and match characteristics. Using data from 14 Grand Slams, serve strength, followed by break point conversion rate and the difference in player ranking points, were found to be the most important predictors of match results. Wilkens[15] applied ML methods to tennis matches and found that average prediction accuracy tends to level off at about 70%. The author also found that, regardless of the model applied, most relevant information was contained in betting odds, and adding other match- or player-specific information did not improve results. The betting odds-implied benchmark and set of models (logistic regression, neural network, random forest,[34] boosting, and SVM[35]) used by Wilkens will be used in the current study to compare with Elo/WElo.

Ratings & Paired Comparison Methods for Tennis Match Result Prediction

In this subsection, studies are reviewed that used rating and paired comparison methods (which are related) to predict tennis match outcomes.

Williams, Liu and Dixon[14] proposed a surface-specific Elo that considers different playing surfaces such as grass, hard and clay courts. Kovalchik[3] compared the performance of several models in predicting 2,395 ATP tennis matches in 2014 and found that Elo performed better than regression-, points- and paired comparison-based models. While Elo ratings have traditionally been updated based on discrete wins/losses in tennis, Elo has been extended to incorporate the margin-of-victory (MoV) by Kovalchik.[4] The author compared four different approaches to incorporate MoV: linear, joint additive, multiplicative, and logistic regression. Although all four approaches were found to outperform standard Elo, the joint additive model was preferred because a simulation study showed it had the most stable variance and smallest bias in terms of player rating differences. Angelini, Candila, and De Angelis[13] noted that standard Elo does not fully take into account current player form and their recent performances. They proposed Weighted Elo (WElo), which places additional weight on the scoreline of the player's last match to account for the "hot-hand" phenomenon (also called "momentum"). With a dataset of over 60,000 ATP and WTA matches, regardless of the training and test periods considered, WElo was found to outperform four competing methods: standard Elo, the Bradley-Terry paired comparison model, the logit regression of Klaasen and Magnus,[26] and the probit regression of Del Corral and Prieto-Rodriguez.[36] The authors also demonstrated that the MoV method of Kovalchik[4] is, in fact, a special case of WElo.

Bradley-Terry[37] paired comparison models are related to Elo because the Elo rating can be expressed as a function of the player's Bradley-Terry strength (subsection 2.1, Coulom[38]). McHale and Morton[39] proposed a Bradley-Terry model that incorporates current and historical player performance measures, and surface-specific parameters. Baker and McHale[40] improved upon this model by proposing a time-varying Bradley-Terry model that improves the ranking of tennis players by allowing players' abilities to vary over time (with player ability following a non-parametric function of time). Another recent study to have used a Bradley-Terry-based approach to forecast tennis match results is that of Fayomi, Majeed, and Algani.[41] A weakness of paired comparison models such as Bradley Terry is that when the number of player pairs is large, sparsity becomes an issue, a problem that some recent studies[42,43] have attempted to address.

Materials & Methods

In this section, Elo and WElo are described in the first subsection, and the experimental framework used in this study, the SRP-CRISP-DM framework, is described, along with how it will be applied, in the following subsection.

Elo and WElo Ratings

Suppose two players, player i and player j, with Elo ratings $R_i(t)$ and $R_j(t)$, respectively, play a match at time t. In standard Elo, the probability that player i beats player j is estimated (in what is known as the Estimation- or E-step) using the logistic curve:

$$p_{i,j}(t) = \frac{1}{1 + 10^{\frac{-(R_i(t) - R_j(t))}{400}}}$$

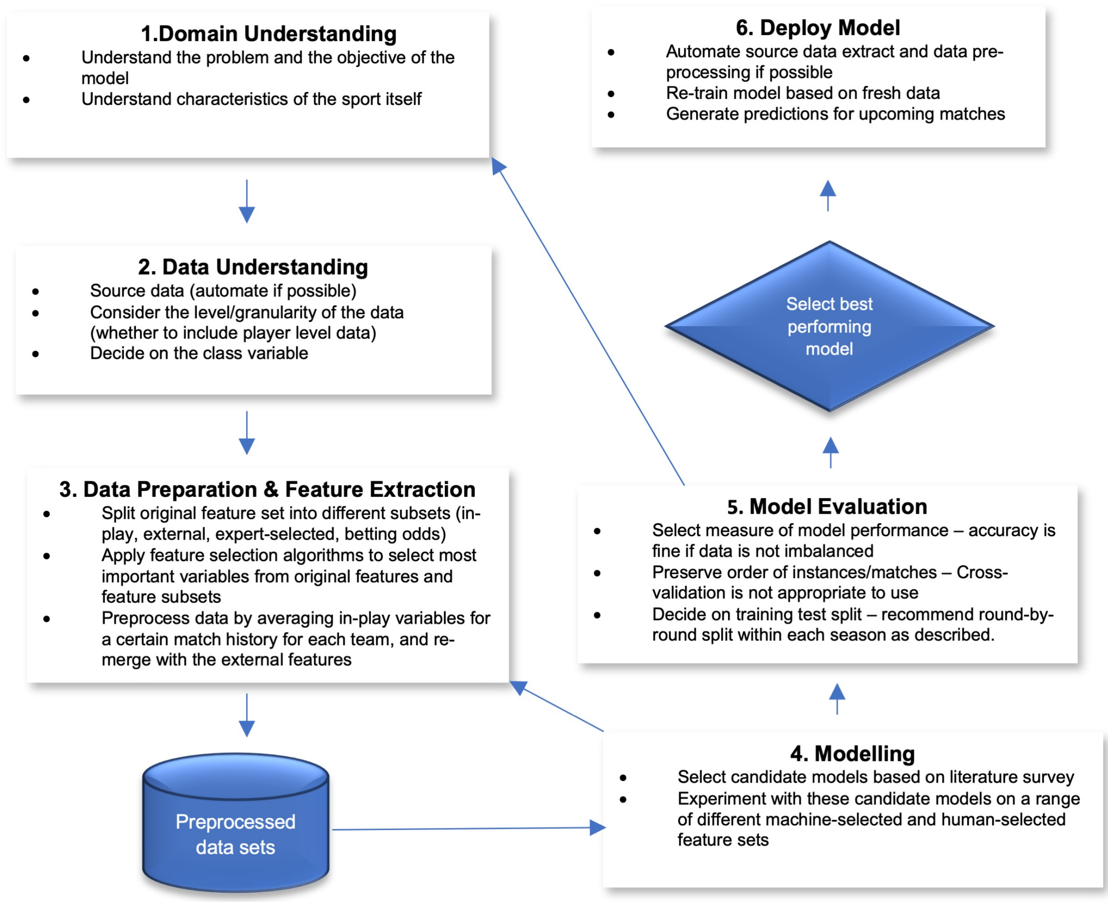
Similarly, the probability that player j beats player i is:

$$p_{j,i}(t) = \frac{1}{1 + 10^{\frac{-(R_j(t) - R_i(t))}{400}}}$$

Because these are probabilities, they add to one, that is, $p_{i,j}(t) + p_{j,i}(t) = 1$. A function representing the actual match result, $A_i(t)$, takes the value 0 if player i loses the match and 1 if player i wins the match, and player i's ratings are updated based on the following update step (U-step):

$$R_i(t+1) = R_i(t) + K_i(t)[A_i(t) - p_{i,j}(t)]$$

Figure 1. Steps of the Sports Result Prediction CRISP-DM (SRP-CRISP-DM) model, which extends the CRISP-DM framework for the specific problems faced in sports match result prediction tasks (two arrows that were not shown in the original paper[24] have been added to the figure).



In WElo,[13] player i's WElo rating, $E_i(t)$, is instead updated according to the following U-step:

$$E_i(t+1) = E_i(t) + K_i(t)[A_i(t) - p_{i,j}(t)]f(G_{i,j}(t))$$

where $f(G_{i,j}(t))$ is a function that depends on the number of games (or sets) won by each player in their previous match. Specifically, $f(G_{i,j}(t)) = \frac{N_i(t)}{N_i(t)+N_j(t)}$ if player i won match t and $f(G_{i,j}(t)) = \frac{N_j(t)}{N_i(t)+N_j(t)}$ if player i lost match t.

A fixed value of $K_i(t) = \bar{K} = 32$ has often been used in the literature. However, Kovalchik[3] found that an Elo rating system devised by FiveThirtyEight,[44] with values of $c = 250$, $o = 5$ and $s = 0.4$, in which K decreases with the number of matches a player has played in their career, performed best against several competing methods:

$$K_i(t) = \frac{c}{(M_i(t)+o)^s} = \frac{250}{(M_i(t)+5)^{0.4}} \tag{1}$$

where c is a constant, $M_i(t)$ is the number of matches in the dataset up to time t that have involved player i, o is a small offset that avoids very large values when $M_i(t)$ is small, and s is a curve shape-altering parameter. The "Kovalchik" K-value in Equation 1, with the above-mentioned FiveThirtyEight values, which is the default option in the WElo R package,[45] is used in the current study.

Experimental Framework: SRP-CRISP-DM

The SRP-CRISP-DM framework[24] extends the original CRISP-DM framework[46] for the specific issues faced when forecasting match results in sports. Like CRISP-DM, the SRP-CRISM-DM framework consists of six steps, which are illustrated in Figure 1.

In this study, the SRP-CRISP-DM framework steps are applied as follows.

1. Domain Understanding: In the current study, the objective of Elo/WElo and the ML models is to predict the results of tennis matches. Although one use case of match result prediction models is to devise profitable betting strategies through betting on sports matches, devising such strategies is not the focus of the current study.

2. Data Understanding:

Data Collection. Elo/WElo and the ML models were applied to the ATP men's dataset used by Angelini, Candila, and De Angelis,[13] which was originally sourced from tennis-data.co.uk and contains ATP matches from 5 July 2005 to 22 November 2020. The ATP data were downloaded as an RData file from the "Appendix C. Supplementary materials" section of the paper.[13] An R script was written that cleans the data (using the 'clean()' function in the WElo R package[45]) and then converts and exports it to a CSV file. The data cleaning reduced the number of matches from 38,868 to 33,976.

3. Data Preparation & Feature Extraction:

Data Preparation. Due to the structure of the data used for the Elo and WElo calculations in the initial data, the dataset needed to be transformed to apply the ML models. In particular, the initial dataset was structured such that each instance represents a specific match, with the winner always appearing in the column preceding the loser. The dataset with the winner always preceding the loser meant that a class/target variable with win/loss could not be created, which is necessary to apply supervised ML models. Therefore, a transformed dataset for ML was created so that the player order was arranged whereby, for each match, Player 1 "P1" and Player 2 "P2" columns were created, where P1 is the first player in terms of alphabetical order based on their surname. The target variable was defined as the match result from Player 2's perspective, that is, whether Player 2 won or lost against Player 1. Alphabetical ordering, based on surname, was designed to produce a dataset that was random and also balanced in terms of the target variable. Indeed, in the transformed dataset, the number of matches belonging to each of "P2 win" and "P2 loss" was split nearly 50:50, with 16,841 P2 wins (49.6%) and 17,135 P2 losses (50.4%). The features for ML were then created to be the differences between Player 1 and Player 2 (Player 1 minus Player 2).

Feature Extraction. Betting odds have proven to be useful for match result prediction in sports, even when used as the only feature in a model.[47] However, it has been pointed out that if the purpose of the model is to generate profit through betting strategies, betting odds should not be included as a model feature if one wants to "beat the house".[48] Because, as mentioned above, our purpose here is purely match result prediction, that is, profitable betting strategies are not attempting to be generated, it is considered appropriate to include betting odds as a model feature. As shown in Table 1, the candidate features were grouped into four types: in-play features (derived from events that occurred within matches), external features, and date and target (class) features. There are some potentially redundant features in the dataset. For example, the difference in Bet365 odds (B365OddsDiff) is correlated with AvgOddsDiff, the difference in average odds across 11 betting companies including Bet365 (Stoke-on-Trent, United Kingdom). In other words, Bet365 is one of the betting companies over which the average odds are calculated. Several ranking and ATP points features were also engineered: the absolute difference in the ATP points and rankings, the percentage difference in the ATP points and rankings, and the difference in ATP points and rankings normalized to be between -1 and 1. It was subsequently realized that the in-play features could not be used for match result forecasting, because the values of these features are, of course, not known until after matches have been played (unless some form of historical averaging process was applied to take the average of each of these in-play features across a certain number of historical matches, which, for simplicity, was not undertaken in this study). Nonetheless, how the values of the six in-play features differ between Player 2 wins and losses is of interest from a practical perspective and is presented in the supplemental material (Table 7). The remaining (external) features were used to predict the target variable because the date feature was only used for splitting the dataset into training and test sets.

Feature Selection. Five different feature selection methods from WEKA (correlation, chi-squared, information gain ratio, reliefF,[49] and symmetrical uncertainty attribute evaluation) were applied to the remaining features and their average rank was taken (Table 2). As mentioned, because the average betting odds difference (AvgOddsDiff) was calculated using the average odds taken across 11 companies including Bet365, it is correlated with the Bet365 betting odds difference (B365OddsDiff). As a result, only AvgOddsDiff, which had a higher average rank than B365OddsDiff, was ultimately used in the final ML models. Likewise, only one of the features derived from ATP rankings or ATP points (PtsDiff, RankDiff, PtsDiffPerc, RankDiffPerc, PtsDiffNorm, and RankDiffNorm) was used in the final ML models. In particular, because PtsDiffNorm, the ATP points difference normalized to be between -1 and 1, had the highest average rank among these six features, it was selected for the final ML models. The final set of features used in the ML models was: AvgOddsDiff, P1 Seed, P1 Entry, SameHand, HeightDiff, SameCountry, AgeDiff, and PtsDiffNorm.

Table 1. Features in the initial transformed dataset, categorized into in-play and external features. *The date variable was used only for creating the training-test splits.

Feature	Description	Type
AcePercDiff	Difference in the percentage of points in the match that were aces	In play
DFPercDiff	Difference in the percentage of points in the match that were double faults	In play
FirstServePercDiff	Difference in first serve success percentage	In play
FirstServePtWinPerc	Difference in the percentage of points won when the first serve was successful	In play
PtsWonOnScndServePerServiceGameDiff	Difference in the points won on second serve per service game	In play
BreakPtSavePerc	Difference in the percentage of break-points saved	In play
Date*	Date of the match (used for the training-test split but not as a model feature)	Date
RankDiff	Difference in players' ATP ranks	External
PtsDiff	Difference in players' ATP competition points	External
RankDiffPerc	Percentage difference in players' ATP ranks	External
PtsDiffPerc	Percentage difference in players' ATP competition points	External
RankDiffNorm	Difference in players' ATP ranks, normalized to be between -1 and 1	External
PtsDiffNorm	Difference in players' ATP competition points, normalized to be between -1 and 1	External
B365OddsDiff	Difference in Bet365 odds	External
AvgOddsDiff	Difference in the average odds across 11 betting companies including Bet365 (Bet365, Bet&Win, Centrebet, Expekt, Ladbrokes, Gamebookers, Interwetten, Pinnacles Sports, Sportingbet, Stan James, and Unibet)	External
P1_Seed	Whether P1 was seeded for the tournament, and if so, what seed they were	External
P1_Entry	Whether P1 had special entry to the tournament, for example, whether they were a qualifier or wild card	External
SameHand	Whether P1 plays with the same hand as P2	External
HeightDiff	Difference in players' heights	External
SameCountry	Whether the players are from the same country	External
AgeDiff	Difference in players' ages (at the time the match was played)	External
P2_Result	Match result from player 2's perspective (that is, "W" if P2 beat P1, "L" if P2 lost to P1)	Target

4. Modelling:

The ML models used in the recent study of Wilkens[15] were compared with Elo/WElo: namely, logistic regression, ANN, random forest, boosting, and SVM. For the boosting method, ADTrees were employed. All models were applied in WEKA 3.9.6 with their default parameters.

Table 2. Feature rank in terms of correlation, chi-squared, information gain ratio, symmetrical uncertainty, and ReliefF (all available in WEKA 3.9.6), along with the average rank across these different methods.

Feature	CorrRank	ChiSqRank	GainRatioRank	SymmURank	ReliefFRank	AvgRank
AvgOddsDiff	1	2	1	1	1	1.2
B365OddsDiff	2	1	2	2	4	2.2
PtsDiffNorm	3	5	5	5	2	4
PtsDiff	4	6	6	6	3	5
RankDiffPerc	8	3	3	3	9	5.2
PtsDiffPerc	7	4	4	4	10	5.8
RankDiffNorm	5	7	7	7	8	6.8
RankDiff	6	8	8	8	7	7.4
P1_Seed	9	9	9	9	11	9.4
HeightDiff	11	11	11	11	6	10
P1_Entry	10	10	10	10	14	10.8
SameHand	12	12	12	12	12	12
AgeDiff	13	14	14	14	5	12
SameCountry	14	13	13	13	13	13.2

The ML models were compared with both Elo and WElo, which are available in the WElo R package.[45] The R script used to apply Elo and WElo is available on GitHub (see the supplemental material).

5. Model Evaluation:

Training-Test Splits. Match result prediction models are often built by training them on a certain number of historical matches and then predicting current or future matches. Because the first year of ATP data (2005) did not include a full year of data, it was excluded from the analysis. The year 2006 was chosen as the initial training set, and one year was incrementally added to this training set and used to predict the subsequent year. That is, the ML models were trained on 2006 and tested on 2007, trained on 2006-2007 and tested on 2008, trained on 2006-2008 and tested on 2009, and so on. Finally, the ML models were trained on 2006-2019 and tested on 2020. Therefore, there were a total of 14 training-test splits. Standard cross-validation randomly shuffles instances; therefore, it cannot be used in match result prediction because future matches will erroneously be used to predict past matches. An option in WEKA was specified to ensure that the temporal order of instances was maintained to ensure only historical matches are used to predict future matches (specifically, the "Preserve order for % Split" option was checked in the "Classifier Evaluation options" in the WEKA workbench under "test options" then "More options...").

Model Evaluation Metric. The imbalance ratio in the dataset was 0.983, which indicates that the size of the minority class was very similar to the size of the majority class. Because the dataset was not unbalanced, accuracy was deemed an adequate measure of model performance.

6. Deployment:

In this study, the models are essentially static in the sense that they do not need to be deployed in any form of production environment and re-run on fresh data.

Results & Discussion

The results of the accuracies of the ML models and Elo/WElo for the 14 training-test splits are shown in Table 3. Also shown in the table are the test set predictions derived purely from the betting odds, that is, based on the rule:

```
if AvgOddsDiff < 0 then P2 result = "L";
else P2 result = "W"
```

It should be noted that, of the 2,350 instances in 2010, 824 (35%) had null values for the AvgOddsDiff feature, and this high proportion of null values in the AvgOddsDiff feature reduced the accuracies achieved by some models (ADTree and LR) in that year. The results show that ADTree achieved the highest accuracy in seven of the 14 test sets. Predictions derived from betting odds achieved the best accuracy in six of the 14 test datasets. Logistic regression achieved the highest accuracy in one of the test sets (2016). The betting odds and ADTree achieved the equal-highest accuracy performance (72.7%) across all test sets. The results obtained are broadly consistent with past studies, for example, that by Wilkens,[15] who found that regardless

Table 3. Accuracy results (%) of the ML models and Elo/WElo.

Train	Test	#Train	#Test	Train%	Test%	LR	ANN	SVM	RF	ADT	Elo	WElo	Odds
2006	2007	2,306	2,368	49%	51%	70.9	65.1	63.7	68.9	71.0	67.3	66.4	**71.5**
2006-2007	2008	4,674	2,242	68%	32%	70.2	65.2	63.9	66.1	**70.9**	67.4	66.7	70.8
2006-2008	2009	6,916	2,278	75%	25%	70.3	65.8	66.5	67.8	**71.2**	69.7	69.6	70.8
2006-2009	2010	9,194	2,350	80%	20%	62.2	63.5	64.6	67.4	63.8	68.0	68.6	**70.8**
2006-2010	2011	11,544	2,345	83%	17%	71.6	69.3	68.6	66.6	71.4	69.3	69.3	**72.3**
2006-2011	2012	13,889	2,324	86%	14%	72.1	69.7	68.1	68.9	72.6	71.2	70.7	**72.7**
2006-2012	2013	16,213	2,316	88%	12%	69.2	67.8	67.6	66.5	68.9	67.3	68.2	**69.4**
2006-2013	2014	18,529	2,250	89%	11%	71.1	68.6	69.6	66.9	70.9	68.9	68.7	**71.3**
2006-2014	2015	20,779	2,285	90%	10%	71.7	66.1	71.4	70.4	**72.7**	69.4	69.3	72.5
2006-2015	2016	23,064	2,297	91%	9%	**71.4**	70.5	70.3	67.7	70.3	68.6	69.0	71.0
2006-2016	2017	25,361	2,307	92%	8%	67.8	65.8	66.7	66.4	**68.0**	67.0	66.5	67.9
2006-2017	2018	27,668	2,008	93%	7%	67.7	66.9	66.1	63.5	**67.9**	63.4	64.0	67.6
2006-2018	2019	29,676	2,303	93%	7%	66.8	66.2	65.5	64.6	**67.3**	63.7	64.0	66.9
2006-2019	2020	31,979	1,033	97%	3%	69.4	67.5	69.6	64.9	**69.8**	63.6	64.1	69.7
						69.5	67.0	67.3	66.9	69.8	67.5	67.5	**70.4**

Note: The highest accuracy for each training-test split is shown in bold, as is the highest average accuracy across all training-test splits in the bottom row.

of the features, models and approaches used, accuracy generally reaches around 70%, and it is difficult to increase beyond that level. The default number of boosting iterations in WEKA, each of which creates an additional ADTree branch, is 10. The ADTree that achieved the equal-highest accuracy across the test sets (72.7%; training set 2006-2014, test set 2015), shown in Figure 2, was overly complex since the AvgOddsDiff feature was present in multiple branches.

However, for the 2006-2014 training 2015 test split, even when the number of boosting iterations was set to one, that is, the ADTree has only one branch (Figure 3), the same accuracy as the 10-branch ADTree was achieved. Since the sum of each predictor node in the ADTree determines the class in which a fresh instance is to be classified, the decision rules that can be derived from this ADTree are:

```
if AvgOddsDiff>=-0.431, 0.014+(-0.401)=-0.387<0
    => P2 result = "W";
else if AvgOddsDiff<-0.431, 0.014+0.509=0.523>0
    => P2 result = "L"
```

Likewise, Figure 4 shows an ADTree with one boosting iteration, trained on 2006-2008 matches and tested on 2009 matches, which achieved 71.2% accuracy, which was the same as the 10-boosting-iteration ADTree and higher than the 70.8% accuracy that was derived from the betting odds. The decision rules that can be derived from this ADTree are:

```
if AvgOddsDiff>=0.328, 0.023+(-0.515)=-0.492<0
    => P2 result = "W";
else if AvgOddsDiff<0.328, 0.023+0.389=0.412>0
    => P2 result = "L"
```

Figure 2. ADTree with 10 boosting iterations, trained on 2006-2014 matches and tested on 2015 matches, which achieved the equal-highest accuracy of 72.7%.

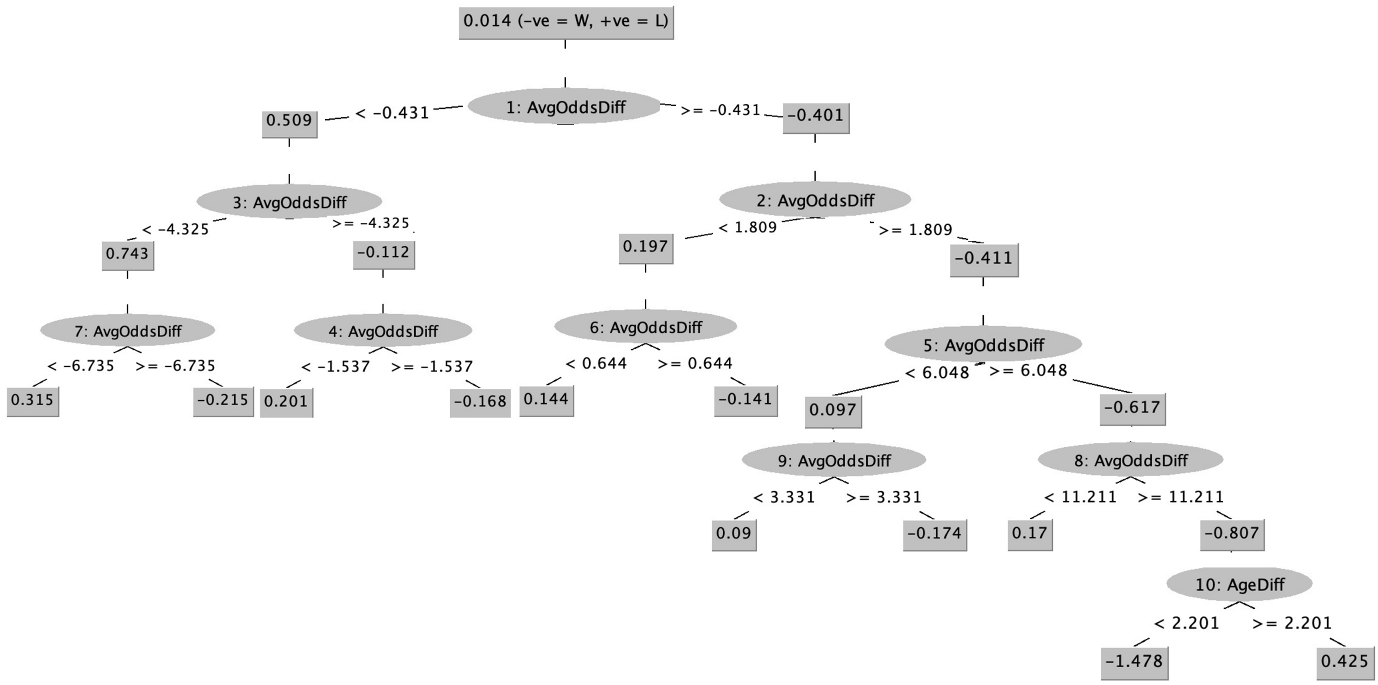
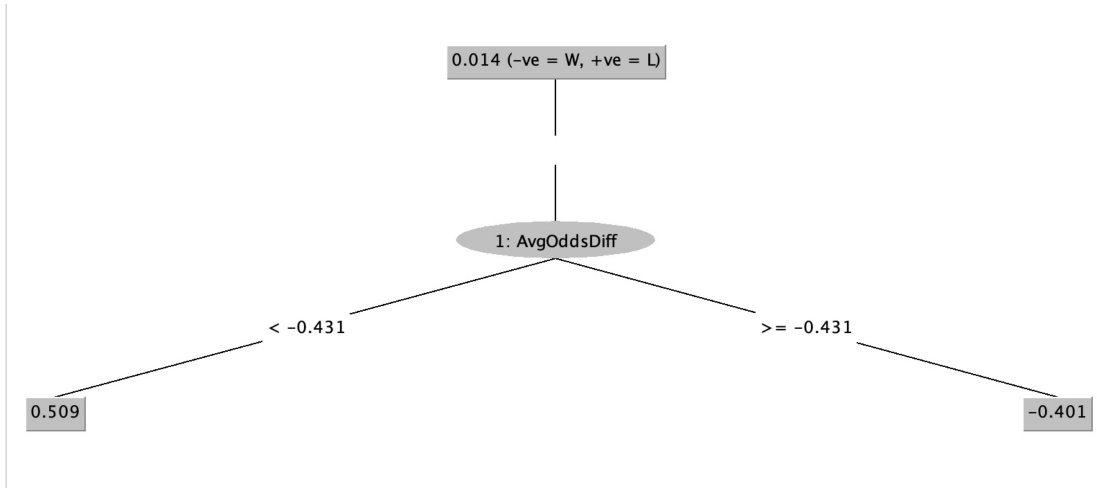


Figure 3. ADTree with one boosting iteration, trained on 2006-2014 matches and tested on 2015 matches, which also achieved 72.7% accuracy.



In fact, the one-branch ADTrees provided the same accuracy performance as the ten-branch ADTrees in all but one test set, the 2020 test set, where the one-branch ADTree achieved a slightly lower accuracy of 68.3%. While predictions derived from betting odds use an AvgOddsDiff threshold of zero on the test set to determine the target variable value, the trained ADTrees allow for this threshold to vary across test sets. The AvgOddsDiff thresholds for each training-test split, which split the one-branch ADTrees, are shown in Table 4.

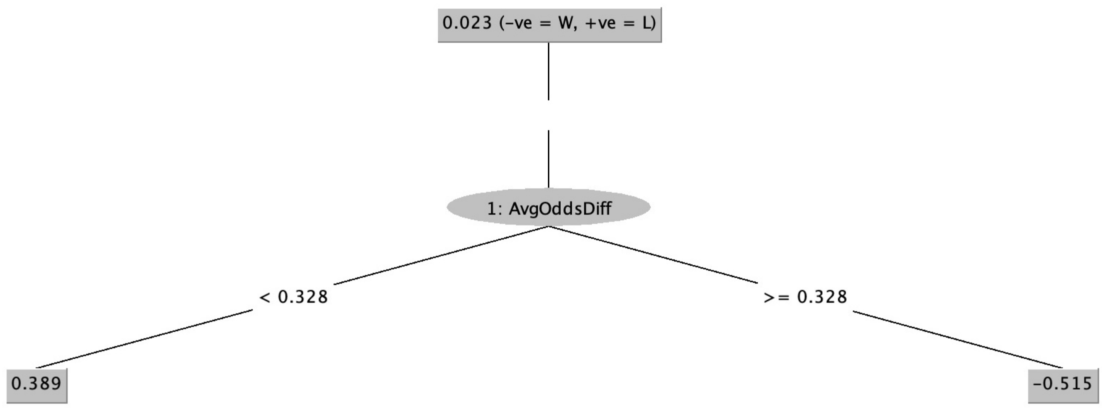
As can be seen, while deriving predictions purely from the betting odds implies a threshold value of zero for AvgOddsDiff, the ADTree allows for this threshold to vary, for example, to -0.431 in the case of the ADTree in Figure 3 and 0.328 in the case of the ADTree in Figure 4.

Table 5 shows the differences in accuracy between each of the ML models and Elo/WElo. The results show that, unlike the ANN, SVM and Random Forest models, the ADTree and Logistic Regression models outperformed both Elo and WElo

Table 4. AvgOddsDiff thresholds from the one-branch ADTrees for each training-test split. All 1-branch ADTrees, apart from this training-test split (where the 1-branch ADTree achieved accuracy of 68.3% versus the 10-branch tree's 69.8%), obtained the same accuracy as the default 10-branch ADTrees.

Train	Test	#Train	#Test	Train%	Test%	ADTree Accuracy (1-branch)	AvgOddsDiff Threshold
2006	2007	2,306	2,368	49%	51%	71.0	0.312
2006-2007	2008	4,674	2,242	68%	32%	70.9	0.328
2006-2008	2009	6,916	2,278	75%	25%	71.2	-0.431
2006-2009	2010	9,194	2,350	80%	20%	63.8	0.328
2006-2010	2011	11,544	2,345	83%	17%	71.4	0.328
2006-2011	2012	13,889	2,324	86%	14%	72.6	0.328
2006-2012	2013	16,213	2,316	88%	12%	68.9	0.328
2006-2013	2014	18,529	2,250	89%	11%	70.9	-0.431
2006-2014	2015	20,779	2,285	90%	10%	72.7	-0.431
2006-2015	2016	23,064	2,297	91%	9%	70.3	-0.431
2006-2016	2017	25,361	2,307	92%	8%	68.0	0.050
2006-2017	2018	27,668	2,008	93%	7%	67.9	-0.117
2006-2018	2019	29,676	2,303	93%	7%	67.3	-0.431
2006-2019	2020	31,979	1,033	97%	3%	68.3	-0.431*

Figure 4. ADTree with one boosting iteration, trained on 2006-2008 matches and tested on 2009 matches, which achieved 71.2% accuracy (the same as the 10-boosting-iteration ADTree, and higher than the 70.8% accuracy derived from the betting odds in the 2009 test set).



on all training-test splits. The respective average differences in accuracy between the ML models and Elo/WElo across all training-test splits were 2.3% and 2.0% for ADTree and Logistic Regression. It is also notable that Elo and WElo ratings tended to outperform the SVM when the amount of training data used was small.

Table 5. Accuracy differences (%) between the ML models and Elo/WElo

Train	Test	#Train	#Test	Train%	Test%	LR-Elo	LR-WElo	ANN-Elo	ANN-WElo	SVM-Elo	SVM-WElo	RF-Elo	RF-WElo	ADT-Elo	ADT-WElo
2006	2007	2,306	2,368	49%	51%	3.6	4.5	-2.2	-1.3	-3.6	-2.8	1.5	2.4	3.7	4.6
2006-2007	2008	4,674	2,242	68%	32%	2.8	3.5	-2.3	-1.6	-3.5	-2.8	-1.3	-0.6	3.4	4.1
2006-2008	2009	6,916	2,278	75%	25%	0.7	0.7	-3.8	-3.8	-3.1	-3.1	-1.9	-1.8	1.5	1.5
2006-2009	2010	9,194	2,350	80%	20%	-5.8	-6.4	-4.6	-5.1	-3.4	-4.0	-0.6	-1.1	-4.2	-4.8
2006-2010	2011	11,544	2,345	83%	17%	2.3	2.3	0.0	0.0	-0.7	-0.7	-2.7	-2.7	2.1	2.1
2006-2011	2012	13,889	2,324	86%	14%	0.9	1.4	-1.5	-1.0	-3.1	-2.6	-2.3	-1.8	1.5	1.9
2006-2012	2013	16,213	2,316	88%	12%	1.9	1.0	0.5	-0.4	0.3	-0.6	-0.8	-1.7	1.6	0.7
2006-2013	2014	18,529	2,250	89%	11%	2.2	2.4	-0.3	-0.1	0.8	1.0	-2.0	-1.7	2.0	2.2
2006-2014	2015	20,779	2,285	90%	10%	2.4	2.5	-3.3	-3.2	2.0	2.1	1.1	1.1	3.3	3.4
2006-2015	2016	23,064	2,297	91%	9%	2.8	2.4	2.0	1.6	1.7	1.3	-0.9	-1.3	1.7	1.3
2006-2016	2017	25,361	2,307	92%	8%	0.8	1.3	-1.2	-0.7	-0.3	0.1	-0.6	-0.1	1.0	1.5
2006-2017	2018	27,668	2,008	93%	7%	4.3	3.7	3.6	3.0	2.8	2.2	0.2	-0.4	4.6	4.0
2006-2018	2019	29,676	2,303	93%	7%	3.1	2.9	2.5	2.2	1.8	1.6	0.9	0.6	3.6	3.3
2006-2019	2020	31,979	1,033	97%	3%	5.8	5.3	3.9	3.4	6.0	5.5	1.3	0.8	6.2	5.7
						2.0	2.0	-0.5	-0.5	-0.2	-0.2	-0.6	-0.6	2.3	2.3

Table 6 shows the differences in accuracy between the ML models and odds-derived predictions on the test sets. The results show that ADTree and Logistic Regression performed comparably to betting odds-derived predictions; however, ANN, SVM, Random Forest, and Elo/WElo performed worse than betting odds.

Table 6. Accuracy differences (%) between the ML models and betting odds-derived predictions on the test sets.

Test	#Train	#Test	Train%	Test%	LR-Odds	ANN-Odds	SVM-Odds	RF-Odds	ADT-Odds	Elo-Odds	WElo-Odds
2007	2,306	2,368	49%	51%	-0.6	-6.4	-7.9	-2.7	-0.5	-4.2	-5.1
2008	4,674	2,242	68%	32%	-0.5	-5.6	-6.9	-4.6	0.1	-3.3	-4.1
2009	6,916	2,278	75%	25%	-0.5	-5.0	-4.3	-3.0	0.4	-1.1	-1.2
2010	9,194	2,350	80%	20%	-8.6	-7.3	-6.2	-3.4	-7.0	-2.8	-2.2
2011	11,544	2,345	83%	17%	-0.8	-3.1	-3.7	-5.7	-0.9	-3.0	-3.0
2012	13,889	2,324	86%	14%	-0.6	-3.0	-4.6	-3.8	0.0	-1.5	-2.0
2013	16,213	2,316	88%	12%	-0.2	-1.6	-1.8	-2.8	-0.5	-2.1	-1.2
2014	18,529	2,250	89%	11%	-0.3	-2.8	-1.7	-4.4	-0.4	-2.4	-2.7
2015	20,779	2,285	90%	10%	-0.7	-6.4	-1.1	-2.1	0.2	-3.1	-3.2
2016	23,064	2,297	91%	9%	0.3	-0.5	-0.8	-3.4	-0.8	-2.5	-2.1
2017	25,361	2,307	92%	8%	-0.1	-2.1	-1.3	-1.5	0.1	-0.9	-1.4
2018	27,668	2,008	93%	7%	0.0	-0.7	-1.5	-4.1	0.3	-4.3	-3.7
2019	29,676	2,303	93%	7%	0.0	-0.7	-1.3	-2.3	0.4	-3.2	-2.9
2020	31,979	1,033	97%	3%	-0.3	-2.2	-0.1	-4.8	0.1	-6.1	-5.6
					-0.9	-3.4	-3.1	-3.5	-0.6	-2.9	-2.9

Conclusions

In this study, the performance in terms of accuracy of five machine learning (ML) models was compared with Elo and WElo ratings in predicting the results of professional men's ATP tennis matches. The differences in performance between the ML models and Elo/WElo and betting odds-derived predictions were also examined.

The results showed that across the 14 training and test sets considered, Logistic Regression and ADTree achieved an average accuracy of 69.5% and 69.8%, respectively, performing comparably to betting odds-derived predictions; however, the ANN, SVM, and Random Forest models – with respective average accuracies of 67%, 67.3%, and 66.9%, and Elo and WElo, which both had an average accuracy of 67.5% – performed worse. Furthermore, the Logistic Regression and ADTree models outperformed Elo and WElo on all training-test splits except for one that was afflicted by a high proportion of null values in the average betting odds difference feature. The ADTree model, by allowing the decision threshold for the average betting odds difference to vary across training-test splits, outperformed odds-derived predictions in half of the training-test splits considered. The dominant model feature was the average (across 11 betting companies) betting odds difference. Since ADTrees retain the interpretability of a decision tree structure, as opposed to commonly used boosted tree methods such as XGBoost and CatBoost, ADTrees may have potential in some areas of sports analytics.

Our results were consistent with recent studies [15,50–52] that have found that accuracy in professional men's tennis match result prediction tends to reach approximately 70%, which is difficult to improve upon and is similar to the accuracy achieved by betting odds alone. This work has added to the growing body of literature suggesting that boosting methods perform well in predicting match results in sports, [51,53,54] and features that contain aggregated information including historical match result results, for example, ratings or betting odds (which can also be considered a rating [55]), can be important.

In future research, historically averaged in-play features could be included as features, the dominant betting odds features could be removed for comparison, and additional (tuned) boosting methods and alternative ratings could be added to the comparison.

Supplemental material

R Code. The R script that was used to apply Elo and WElo is available on GitHub (https://github.com/rorybunker/tennis-elo-vs-ml).

Data Availability. The machine learning models were applied in WEKA and the training/test set (ARFF) files are also available in the above GitHub repository. As mentioned in the Data Collection subsection, the ATP RData file is openly available in the appendix of the online version of the paper of Angelini, Candila, & De Angelis [13] at https://doi.org/10.1016/j.ejor.2021.04.011 under "Supplementary Data S1". This is open data under the CC BY license http://creativecommons.org/licenses/by/4.0/.

In-play features statistical analysis. As mentioned, the in-play variables could not be used for match result prediction since the values of these variables are only known post-match. However, out of practical interest, statistical techniques were used to analyse the differences in these variables between winning and losing outcomes (from Player 2's perspective).

As shown by the Anderson-Darling normality test p-values, which were all very close to zero, the null hypothesis of normality for each of the six features was rejected. Because the two-sample t-test was unable to be employed, the non-parametric Wilcoxon-Mann-Whitney test was used. All p-values from the Wilcoxon-Mann-Whitney test were very close to zero, indicating statistically significant differences between the win and loss class labels for all six variables. Cohen's D effect sizes showed that the difference in the percentage of points won on a first serve ($D = 1.9$), followed by the difference in the percentage of breakpoints saved ($D = 0.91$), followed by the difference in the percentage of first serves that were aces ($D = 0.8$) were the most relevant discriminative features.

Feature ranks for each feature selection technique. Here, in Tables 8 to 12, the feature ranks for each of the feature selection techniques considered are presented, which were used to calculate the average feature ranks in Table 2 and then to select the final feature set.

Table 7. Descriptive statistics between the “W” and “L” target variable labels for the in-play features in the transformed dataset, along with the p-values from Anderson-Darling test for normality (shown in column pA) and Wilcoxon-Mann-Whitney’s test p-values (shown in column pW), as well as Cohen’s D effect sizes (shown in column D). The Wilcoxon-Mann-Whitney test p-values were all <2.2E-16, and are displayed in the table below as ˜0.

target	feature	N	mean	SD	med	min	max	range	skew	kurt	pA	pW	D
L	AcePercDiff	17072	0.03	0.08	0.03	-0.39	0.53	0.91	0.44	1.63	3.7E-24	˜0	0.8
	DFPercDiff	17072	-0.01	0.04	-0.01	-0.23	0.19	0.41	-0.26	1.00	3.7E-24	˜0	0.5
	FirstServePercDiff	17072	0.02	0.11	0.02	-0.43	0.49	0.92	0.07	0.19	7.8E-06	˜0	0.4
	FirstServePtWinPerc	17072	0.11	0.11	0.10	-0.27	0.71	0.98	0.46	0.53	2.9E-04	˜0	1.9
	PtsWonOnScnd ServePerService GameDiff	16832	0.13	0.48	0.12	-3.50	2.5	6.00	0.11	0.39	9.4E-14	˜0	0.5
	BreakPtSavePerc	15409	0.15	0.33	0.16	-1.00	1.00	2.00	-0.18	0.42	1.2E-19	˜0	0.9
W	AcePercDiff	16775	-0.03	0.08	-0.02	-0.47	0.32	0.78	-0.41	1.41			
	DFPercDiff	16775	0.01	0.04	0.01	-0.16	0.27	0.44	0.33	1.36			
	FirstServePercDiff	16775	-0.02	0.11	-0.02	-0.45	0.38	0.83	-0.02	0.18			
	FirstServePtWinPerc	16775	-0.10	0.11	-0.09	-0.67	0.26	0.93	-0.44	0.42			
	PtsWonOnScnd ServePerService GameDiff	16536	-0.12	0.48	-0.11	-2.67	2.00	4.67	-0.08	0.23			
	BreakPtSavePerc	15189	-0.15	0.33	-0.17	-1.00	1.00	2.00	0.19	0.34			

Table 8. Feature ranks in terms of correlation

Rank	Correlation	Feature
1	0.406	AvgOddsDiff
2	0.395	B365OddsDiff
3	0.329	PtsDiffNorm
4	0.329	PtsDiff
5	0.285	RankDiffNorm
6	0.285	RankDiff
7	0.237	PtsDiffPerc
8	0.191	RankDiffPerc
9	0.168	P1_Seed
10	0.096	P1_Entry
11	0.059	HeightDiff
12	0.025	SameHand
13	0.022	AgeDiff
14	0.003	SameCountry

Table 9. Feature ranks in terms of Chi-squared statistic

Rank	ChiSquared	Feature
1	7831.845	B365OddsDiff
2	7475.673	AvgOddsDiff
3	5191.824	RankDiffPerc
4	5058.638	PtsDiffPerc
5	5054.41	PtsDiffNorm
6	5054.41	PtsDiff
7	4066.51	RankDiffNorm
8	4066.51	RankDiff
9	2284.348	P1_Seed
10	368.704	P1_Entry
11	97.871	HeightDiff
12	21.176	SameHand
13	0.376	SameCountry
14	0	AgeDiff

Table 10. Feature ranks in terms of information gain ratio

Rank	GainRatio	Feature
1	0.0516943	AvgOddsDiff
2	0.0505759	B365OddsDiff
3	0.0389762	RankDiffPerc
4	0.0385497	PtsDiffPerc
5	0.0358154	PtsDiffNorm
6	0.0358154	PtsDiff
7	0.0302545	RankDiffNorm
8	0.0302545	RankDiff
9	0.0193729	P1_Seed
10	0.0091313	P1_Entry
11	0.0013614	HeightDiff
12	0.0005692	SameHand
13	0.0000213	SameCountry
14	0	AgeDiff

Table 11. Feature ranks in terms of symmetrical uncertainty

Rank	Feature	SymmU
1	AvgOddsDiff	0.0798728
2	B365OddsDiff	0.0794467
3	RankDiffPerc	0.0587877
4	PtsDiffPerc	0.0577436
5	PtsDiffNorm	0.0546706
6	PtsDiff	0.0546706
7	RankDiffNorm	0.0451663
8	RankDiff	0.0451663
9	P1_Seed	0.0279571
10	P1_Entry	0.0084697
11	HeightDiff	0.001646
12	SameHand	0.0005024
13	SameCountry	0.0000116
14	AgeDiff	0

Table 12. Feature ranks in terms of ReliefF attribute evaluation

Feature	ReliefF	Rank
AvgOddsDiff	0.0048062	1
PtsDiffNorm	0.0040838	2
PtsDiff	0.0040838	3
B365OddsDiff	0.0036873	4
AgeDiff	0.0034504	5
HeightDiff	0.0031599	6
RankDiff	0.0025382	7
RankDiffNorm	0.0025382	8
RankDiffPerc	0.0011072	9
PtsDiffPerc	0.0009	10
P1_Seed	0.0008418	11
SameHand	0.0001295	12
SameCountry	0.0001295	13
P1_Entry	-0.0000294	14

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Declaration of conflicting interests

The authors declare that there are no conflicts of interest.

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